1.2.1.7 Example of type designation. The type designation RVC5NYB103B signifies:

RVC5 - A variable, nonwire-wound resistor of the dimensions and power rating specified (see 3.1).
N - Standard bushing as specified in table I.
Y - Temperature and moisture-resistance characteristic as specified in table II.
D - Shaft 7/8 inch long.
103 - Nominal total resistance value of 10,000 ohms.
B - Linear taper with resistance tolerance of ±10 percent.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-561 - Solder, Silver.
QQ-S-571 - Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-676 - Box, Setup.
PPP-B-636 - Box, Fiberboard.
PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, (For Carton Sealing).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-C-45662 - Calibration System Requirements.

(See supplement 1 for list of applicable detail specifications.)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Detail specifications for individual resistor types. Detail requirements or exceptions applicable to individual types of resistors shall be as specified in the detail specification. In the event of any conflict between requirements of this specification and the detail specifications, the latter shall govern (see 6.1).

3.2 Qualification. Resistors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of the bids (see 4.4 and 6.2).

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the resistors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Operating shaft. All operating shafts shall be of metal construction.

3.4.1.1 Size. The diameter and length of the operating shaft shall be as specified (see 3.1).

3.4.1.2 Slots. The slotted portion shall be as specified (see 3.1). In all cases, operating shafts shall be of such length that the slot depth will not interfere with shaft retainers. The slot shall be so positioned that the longitudinal centerline of the slot is within ±10 degrees of the longitudinal centerline of the contact.

3.4.2 Rotation. The total mechanical rotation and the electrical rotation of resistors shall be as shown on figure 1 and as specified (see 3.1). The total mechanical rotation shall be equal to the total electrical rotation.

3.4.2.1 Clockwise taper. A clockwise taper is a resistance taper in which the resistance varies approximately as shown on figure 2, increasing as the rotation angle increases in a clockwise direction as viewed from the operating shaft, and measured between terminals indicated as 1 and 2 on figure 1.

3.4.3 Contact-arm assembly. Contact pressure on the resistance element shall be maintained uniformly by positive pressure and shall be such as to permit smooth electrical and mechanical control of the resistor over the entire range of electrical rotation (see figure 1). The rotating contact shall have continuous electrical contact with its terminal throughout the entire electrical rotation and shall be insulated from the rotating operating shaft and the resistor housing. The entire contact-arm assembly shall be sufficiently sturdy to preclude the necessity of adjustment during the life of the resistor.

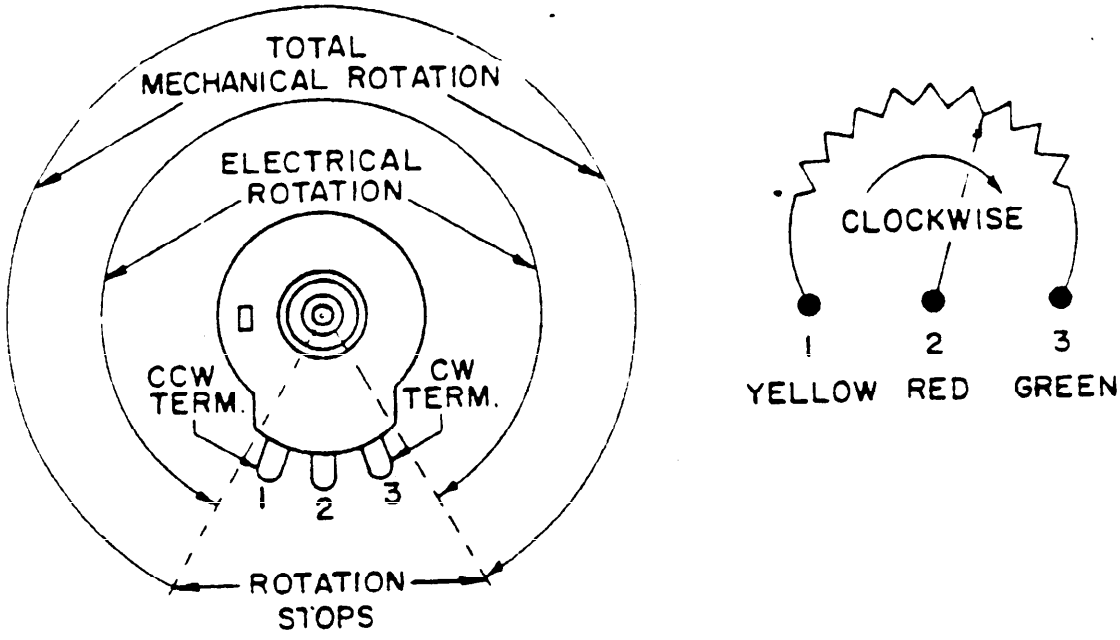


FIGURE 1. Mechanical and electrical rotation.

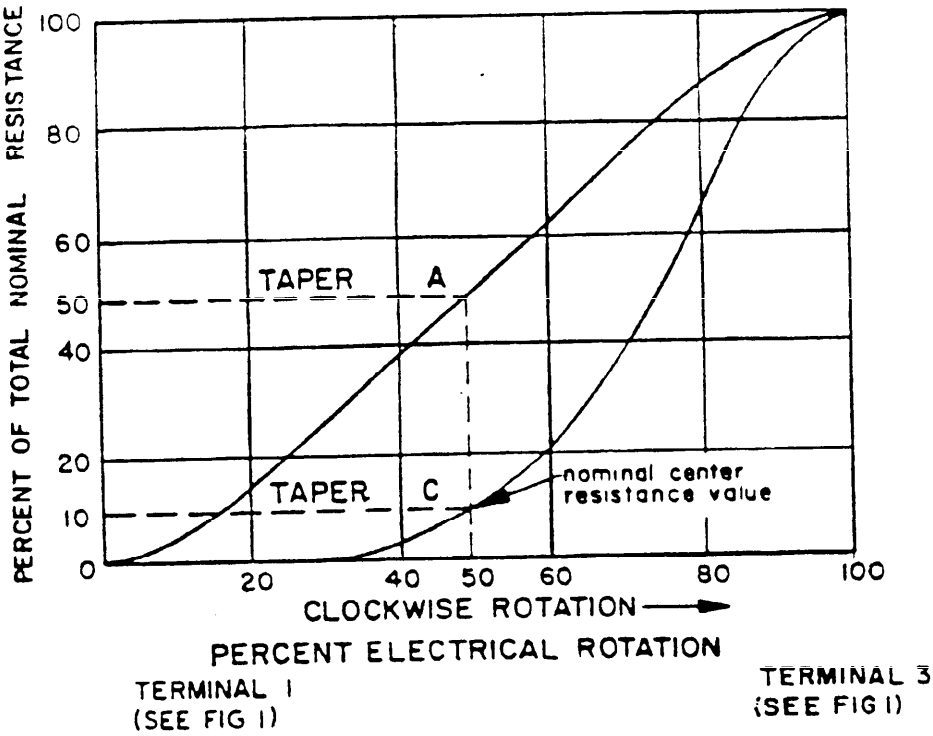


FIGURE 2. Resistance taper.

3.4.3.1 Stops. Stops employed to limit the mechanical rotation of the contact-arm assembly may be a part of, but shall not complete, any electrical circuit.

3.4.4 Terminals. Resistors shall be supplied with external terminal lugs of such size and styles as to permit accommodating and soldering of three 0.032-inch-diameter (AWG size 20, stranded) wires for resistors with 1/4-inch-diameter operating shaft, and three 0.025-inch-diameter (AWG size 22, solid) wires for resistors with 1/8-inch-diameter operating shafts. Terminals shall be suitably treated to facilitate soldering.

3.4.4.1 Terminal designation. Resistor terminals shall be color coded or numbered (see 3.1), and the circuit diagram shall be marked on the back or side of the resistor in a legible manner as shown on figure 1.

3.4.5 Solder. Solder for electrical connections shall be in accordance with QQ-S-561 or QQ-S-571.

3.4.6 Mounting. Resistors shall be mounted by the single-hole method with provision for the nonturn device as specified (see 3.1).

3.4.6.1 Hardware. When specified (see 3.1), each resistor shall be furnished with a corrosion-resistant, internal-tooth lockwasher, a hexagonal mounting nut, and a locking nut if applicable, not assembled on the resistor, unless otherwise specified. For direct Government orders, hardware shall be assembled on the resistor (see 6.1).

3.4.6.2 Standard bushings. When standard bushings are specified, the bushings shall be as shown for the applicable style of resistor (see 3.1).

3.4.6.3 Locking bushings. When locking bushings are specified, the bushings shall be as shown for the applicable style of resistor (see 3.1). The locking device shall be capable of locking the shaft when the resistor is mounted on a panel 0.125-inch thick, maximum.

3.4.6.4 Shaft and panel-sealed bushings. When shaft and panel-sealed bushings are specified, the panel seal may be an integral part of the resistor or may be detachable (see 3.1).

3.4.6.5 Nonturn device. A nonturn device shall be furnished on resistors, which will prevent their rotation with respect to the surface on which they are mounted. The location of the nonturn device and its widest dimension shall be as specified (see 3.1). If the nonturn device is not symmetrical about its vertical axis, it shall be so placed that a plane passed perpendicular to the widest dimension and through its center will pass through the axis of the operating shaft. The nonturn device shall be permanently fastened to the resistor body.

3.4.7 Threaded parts. All threaded parts shall be in accordance with Handbook H28 and as specified (see 3.1).

3.4.7.1 Securing of screws, nuts, and threaded parts. All screw-thread assemblies shall be made resistant to loosening under vibration.

3.4.8 Enclosure. Resistors shall have suitable enclosures for protection against mechanical damage.

3.5 DC resistance (see 4.6.2).

3.5.1 Total resistance. Resistors shall have a total resistance equal to the nominal resistance value plus or minus the resistance tolerance specified (see 1.2.1.5, 1.2.1.6, and 4.6.2.1).

3.5.2 Resistance taper. A linear resistance taper is one having a constant change of resistance with angular rotation. A nonlinear resistance taper is one having a variation or lack of constancy in the change of resistance with angular rotation. The resistance taper shall conform in general shape to the nominal curves shown on figure 2. Resistance measurements shall fall within ±10 percent of the nominal values shown by the curves at the specified angle of 50 percent of electrical rotation (see figure 2 and 4.6.2.2).

3.5.3 Minimum resistance (see 4.6.2.3).

3.5.3.1 Taper-A (linear) resistors. The minimum resistance at either extreme of the mechanical electrical rotation shall not exceed the values listed in table IV.

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for small sample inspection. Unless otherwise specified herein, S-2 inspection shall be used. The acceptable quality level (AQL) shall be 4.0 (percent defective).

4.5.1.3.2 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or order, at the option of the supplier.

4.5.1.4 Group C inspection. Group C inspection shall be performed weekly, monthly, and quarterly and shall consist of the tests specified in table IX, in the order shown. They shall be performed on sample units from lots that have passed all groups A and B inspection.

TABLE IX. Group C inspection.

Test	Number of sample units to be tested		Requirement paragraph	Test paragraph
Weekly				
Subgroup 1				
Low-temperature storage	6	12	3.16	4.6.13
Low-temperature operation			3.17	4.6.14
Temperature cycling			3.18	4.6.15
Subgroup 2				
Resistance to soldering heat	6		3.11	4.6.8
Rotational life			3.12	4.6.9
Monthly				
Resistance-temperature coefficient	6	6	3.9	4.6.6
Moisture-resistance			3.14	4.6.11
Terminal strength			3.15	4.6.12
Quarterly				
Subgroup 1				
Shock, medium impact	6	16	3.20	4.6.17
Vibration, high frequency			3.21	4.6.18
Salt spray (corrosion)			3.22	4.6.19
Subgroup 2				
Life	10		3.13	4.6.10
Subgroup 3				
Solderability	10		3.10	4.6.7
Subgroup 4				
High-temperature exposure	10		3.19	4.6.16

4.5.1.4.1 Sampling plan. Sample units in each style currently in production shall be selected. The number of sample units to be inspected shall be as specified in table IX. Separate samples of the size required by table IX shall be used for each subgroup listed. In the weekly and monthly tests, one defective unit for each subgroup shall be allowed. In the quarterly tests, one defective unit shall be allowed for subgroup 1 tests. One defective unit shall be allowed for quarterly subgroups 2, 3 and 4 combined.

4.5.1.4.2 Disposition of sample units. Sample units subjected to group C inspection shall not be delivered on the contract or order.

4.5.1.4.3 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspection may be reinstituted; however, final

corrective action was successful. In the event of failure [illegible], information concerning the failure and the corrective action taken shall be furnished to the contracting officer.

4.5.2 Inspection of preparation for delivery. Sample packages and packs shall be selected and inspected in accordance with the schedule of acceptance tests and visual inspection aids of MIL-P-116, to verify conformance with the requirements in section 5 of this specification.

4.5.3 Retention of qualification. To retain qualification, the supplier shall forward at 6-month intervals, to the qualifying activity, a summary of the results of groups A and B tests, indicating as a minimum the number of lots which passed and the number which failed, and a summary of the results of group C tests, including the number and type of any part failures. The summary shall include those tests performed during that 6-month period. If the summary of the tests results indicates nonconformance with specification requirements, action shall be taken to remove the failing product from the qualified products list. Failure to submit the summary shall result in loss of qualification for that product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 6-month period that the inspection data indicates failure of the qualified product to meet the requirements of the specification. In the event no product has been submitted under the specification during the 6-month period, a statement indicating the condition shall be forwarded to the qualifying activity in the manner indicated above.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination. Resistors shall be examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.3 to 3.4.8 inclusive, and 3.23 to 3.24.2 inclusive).

4.6.2 DC resistance (see 3.5). Resistors shall be tested in accordance with method 303 of MIL-STD-202. The following details shall apply:

(a) Limit of error of measuring apparatus - Shall not exceed ±0.5 percent for qualification inspection nor ±1 percent for quality conformance inspection.
(b) Voltage - As specified in table X.
Voltages shall be those actually impressed across the resistors. The voltage selected shall be used for all tests and all resistors having the same nominal total resistance value shall be measured with the same test voltage.

TABLE X. Voltage ranges.

Nominal total resistance	Maximum test potential
Ohms	Volts
100 to 999 incl	2
1,000 to 9,999 incl	4
10,000 to 99,999 incl	15
0.1 megohm and over	40

4.6.2.1 Total resistance (see 3.5.1). The total resistance of resistors shall be measured between the terminals of the resistance element with the contact arm against either stop for taper-A and against the stop at the low-resistance end for taper-C. The resistance value obtained shall be compared with the specified nominal total resistance.

4.6.2.2 Resistance taper (see 3.5.2). Following the measurement of total resistance, a resistance measurement shall be made at 50 percent of electrical rotation. The contact arm may be varied ±3 percent of total electrical rotation from the specified angle of 50-percent rotation to meet the resistance tolerance of ±10 percent.

4.6.2.3 Minimum resistance (see 3.5.3). The contact arm shall be rotated to its extreme counterclockwise limit of mechanical electrical rotation. With the arm in this position, the resistance between the counterclockwise terminal and the rotating contact terminal shall be measured. The contact arm shall then be rotated to its extreme clockwise limit of mechanical electrical rotation. With the arm in this position, the resistance between the clockwise terminal and the rotating contact terminal shall be measured.

4.6.3 Torque (see 3.6).

4.6.3.1 Operating. The torque required to rotate the contact arm of the resistance element shall be determined throughout the entire range of mechanical rotation by the torque-wrench method.

4.6.3.2 Stop. Resistors shall be mounted on a metal panel by their normal mounting means. The contact arm shall then be rotated to each extreme limit of mechanical rotation, and the specified torque (see 3.1), applied through the operating shaft to the stops.

4.6.3.3 Locking. Resistors shall be mounted on a metal panel by their normal mounting means and the locking device tightened with a torque not greater than that specified (see 3.1). After the locking device is tightened, the specified torque shall be applied to the shaft.

4.6.4 Dielectric withstanding voltage (see 3.7).

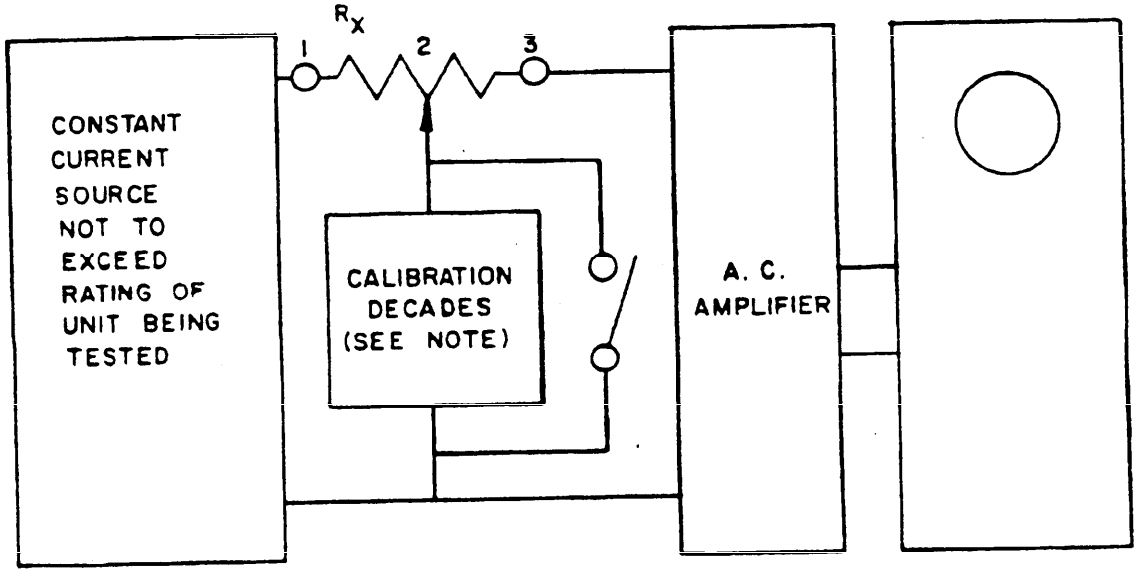
4.6.4.1 At atmospheric pressure. Resistors shall be tested in accordance with method 301 of MIL-STD-202. The following details and exception shall apply:

(a) Special preparation - Resistors shall be mounted on metal plates of sufficient size to extend beyond the resistor extremities, and in such a manner that measurements can be made between the terminals tied together and any other external metal parts.
(b) Magnitude of test voltage - 900 volts root mean square (rms) (750 volts for RVC6).
(c) Nature of potential - From an alternating-current (ac) supply at commercial-line frequency and waveform.
(d) Points of application of test voltage - Between the terminals connected together and all external metal portions of the resistors and metal-mounting plate.
(e) Examinations and measurements - During the tests, the leakage current shall not exceed 1 milliampere, and the resistors examined for evidence of arcing and breakdown. At the conclusion of the test, resistors shall be examined for evidence of damage.

4.6.4.2 At reduced barometric pressure. Resistors shall be tested in accordance with method 105 of MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting - As specified in 4.6.4.1(a).
(b) Test condition - C.
(c) Period of time at reduced pressure prior to application of potential - 1 minute.
(d) Tests during subjection to reduced pressure - The specified potential (see 3.1), from an ac supply at commercial-line frequency and waveform shall be applied for 1 minute.
(e) Points of application of test voltage - As specified in 4.6.4.1(d).
(f) Examinations and measurements - As specified in 4.6.4.1(e).

4.6.5 Contact-resistance variation (see 3.8). Units shall be mounted by their normal mounting means (see 3.1). Contact-resistance variation shall be measured in accordance with figure 3 or its equivalent. During this test the shaft shall be rotated in both directions through 90 percent of the mechanical travel at the rate of 1 cycle for 5 seconds minimum to 1 minute maximum. Contact-resistance variation shall be defined as any abrupt change from one contact resistance level to another, exclusive of the roll-on roll-off points where the contact arm moves from the termination to the resistance element and at rotation reversal points.



RX - Test specimen.
Test bandwidth - 100 Hz to 50 KHZ
Minimum input impedance - At least 10 times nominal resistance being tested.

NOTE:
At the calibration of the decade terminals 1 and 2 shall be coincident.
Calibration decade is to be set for the contact-resistance variation (CRV) level of the specific nominal resistance being tested.
Switch out decades when measurements are taken.

FIGURE 3. Contact resistance variation.

4.6.6 Resistance-temperature coefficient (see 3.9). Resistors shall be tested in accordance with method 304 of MIL-STD-202. The following details and exceptions shall apply:

(a) Reference temperature - Room ambient temperature.
(b) Test temperature - In accordance with table XI.
(c) Accuracy of temperature measurement - Resistors shall be maintained for 30 to 45 minutes at each of the test temperatures listed in table XI. The temperature tolerance of table XI shall be maintained on the established test temperatures.
(d) Units shall be mounted by their normal mounting means (see 3.1).

TABLE XI. Resistance-temperature coefficient.

Sequence	Temperature	
	Qualification inspection	Group C inspection
	°C ±2°C	°C ±2°C
1	Room temperature[1/]	Room temperature[1/]
2	-15	-65
3	-65	Room temperature[1/]
4	Room temperature[1/]	125
5	+65	---
6	125	175
7	150	---
8	175	---

1/ This temperature shall be considered the reference temperature for each of the succeeding temperatures.

4.6.7 Solderability (see 3.10). Resistors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

(a) All three terminals shall be tested.
(b) No special preparation of leads is required.
(c) Aging shall apply.
(d) Terminals shall be dipped within 0.125 inch of body. All three terminals shall be dipped simultaneously whenever possible. When configuration is such that terminals cannot be dipped simultaneously, the 0.125-inch shall apply to the terminal on whichever it occurs first when all three terminals are dipped in an orientation which most nearly meets the above requirement.
(e) Evaluation for tab terminals shall apply.

4.6.8 Resistance to soldering heat (see 3.11). Resistors shall be tested in accordance with method 210 of MIL-STD-202. The following details shall apply:

(a) Measurement before test - DC resistance shall be measured as specified in 4.6.2.
(b) Test condition - A.
(c) Depth of the immersion in the molten solder - To a point within 1/8 inch to 3/16 inch from the resistor body. Immersion procedure shall be as specified in 4.6.7(d).
(d) Measurement after test - Twenty-four ±4 hours after completion of test, the dc resistance shall be measured as specified in 4.6.2. Resistors shall be examined for evidence of mechanical damage.

4.6.9 Rotational life (see 3.12).

4.6.9.1 Mounting. Resistors shall be mounted as specified in 4.6.10(a). Care shall be taken to insure that side thrust is minimized by the proper alinement of the resistor shafts with the drive shafts. The resistors shall be ganged in pairs and each pair shall be connected in series as shown on figure 4, so that a nominally constant current flows through the resistor, irrespective of the contact arm position during the cyclings of the shafts. When performing this test on locking-bushing types, the locking nut shall be removed. If it is desired to rotate paired resistors face to face on the same drive shaft, the appropriate electrical connections equivalent to figure 4 shall be used.

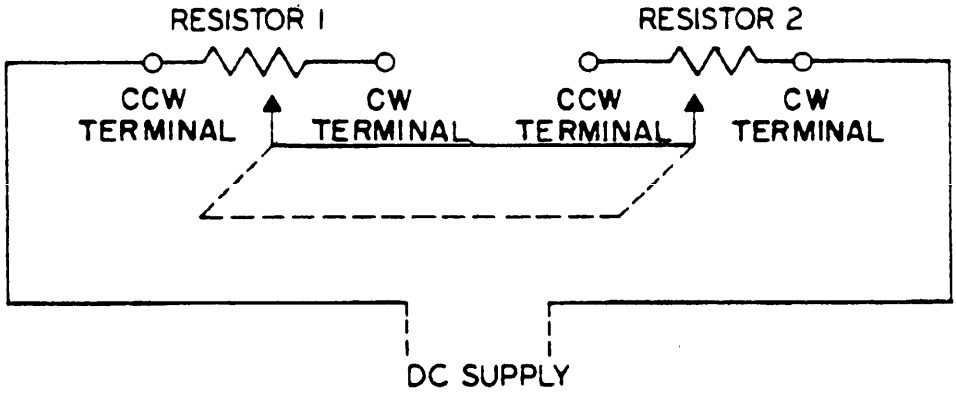


FIGURE 4. Rotational life.

4.6.9.2 Rotation. Total resistance shall be measured as specified in 4.6.2.1. A direct current (dc) potential equivalent to that required to dissipate rated-wattage across the entire resistive element of resistors having the same nominal total resistance as those under test shall be applied as shown on figure 4. Resistor shafts shall then be continuously cycled through not less than 90 percent of the total mechanical rotation, at a rate of 10 ±2 cycles-per-minute for not less than 4,000 cycles-per-day for a total of 50,000 cycles (total cycles shall be 1,000 for locking bushings). Total resistance shall be measured at the following intervals:

Bushings L and T	Bushings S and N
0 cycle	0 cycle
1,000 cycles	25,000 cycles
---	50,000 cycles

Upon completion of the cycling portion, contact resistance variation shall be measured as specified in 4.6.5. Resistors shall be examined for mechanical damage. At the option of the supplier, the sample units may be rotated at a rate of 10 cycles to 50 cycles-per-minute for quality conformance inspection.

4.6.10 Life (see 3.13). Resistors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting - By means of their mounting bushings in the center of a vertical No. 16 gage, steel panel, 4 inches square in still air. Resistors and plates shall be oriented so that the longitudinal axis of the shaft is horizontal, radial terminals downward, axial terminals with wiper terminal above end terminals. The plate shall be in an approximately vertical plane with edges of plate horizontal and vertical. Still air is surrounding air with no circulation other than that created by the heat of the resistor being operated. No shielding shall be located closer than 12 inches from the panel.

(b) Test temperature and tolerance - 125° ±2°C.

(c) Initial measurement - Minimum resistance at $25^{\circ}\ ^{+10^{\circ}}_{-5^{\circ}}$ C (see 4.6.2.3). Following this measurement, the ambient temperature shall be held at the test temperature of 24 ±4 hours without load. At the end of this period, total resistance shall be measured. This measurement shall be the initial test measurement which shall be used as a reference for all future measurements of total resistance.

(d) Operating conditions - Rated direct-current (dc) continuous working voltage as specified (see 3.1), shall be applied intermittently 1-1/2 hours "on" and 1/2 hour "off" for a total of 1,000 hours through the contact arm located against either stop and one resistance element terminal for taper-A, and against the stop at the low resistance end and one resistance element terminal for taper-C, so that rated wattage is dissipated across the entire resistance element. Adequate precautions shall be taken to maintain constant voltage on resistors under test.

(e) Test condition - D.

(f) Measurements during test - Resistance measurements shall be made at the end of the 1/2 hour "off" periods after 100 ±8, 200 ±8, 500 $^{+48}_{-0}$ and 1,000 $^{+48}_{-0}$ hours have elapsed and compared to the similar reading taken in (c).

(g) Measurements after test - The resistors shall be returned to an ambient temperature of 25° $^{+10°}_{-5}$ C, and the minimum resistance shall be measured within not less than 2 hours nor more than 4 hours. Contact-resistance variation shall also be measured as specified in 4.6.5.

4.6.11 Moisture resistance (see 3.14). Resistors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - As specified in 4.6.10(a). Vibration shall be performed either in a vertical or horizontal plane.

(b) Initial measurements - Total resistance shall be measured between the left terminal and the contact-arm terminal with the operating shaft rotated to the maximum clockwise position (see figure 1).

(c) Load - During steps 1 and 4, rated (100 percent) wattage or maximum rated dc voltage, whichever is smaller, shall be applied for the first 2 hours of each step.

(d) Method of applying load voltage - Across the left terminal and the contact-arm terminal, with the operating shaft rotated to the maximum clockwise position (see figure 1).

(e) Final measurements - At the end of the final cycle and while the resistors are still in the humidity chamber at the high-humidity condition, total resistance shall be measured as specified in (b). Following this measurement, the humidity chamber may be opened in order to make the necessary electrical connections to perform the dielectric-strength and insulation-resistance tests. The chamber shall then be returned to the high-humidity condition and maintained at this condition for at least 1 hour prior to these tests. Dielectric withstanding voltage shall be tested as specified in 4.6.4.1 except that the operating shaft shall not be rotated during the test, electrification time shall be 2 minutes, and the test potential applied shall be 500 volts. Insulation resistance shall then be measured between the resistor terminals tied together and the mounting bushing, using a dc potential of 500 volts. The subsequent 4-hour to 24-hour conditioning period and measurements do not apply.

4.6.12 Terminal strength (see 3.15). Resistors shall be tested in accordance with method 211 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition - A.
Applied force - Two pounds applied gradually to each terminal, one at a time, in a direction parallel to the major axis of the terminal.

(b) Examination after test - Resistors shall be tested for electrical continuity and examined for evidence of mechanical damage.

4.6.13 Low-temperature storage (see 3.16). The total resistance shall be measured at 25° ±5°C. Within 1 hour after this measurement, resistors shall be placed in a chamber at a temperature of 25° ±5°C. The temperature in this chamber shall be lowered to -65° ±2°C within a minimum period of 3 hours. Twenty-four hours after the resistors have reached this temperature, the temperature of the chamber shall be gradually raised to 25° ±5°C within a maximum period of 8 hours. Resistors shall then be removed from the chamber and maintained at 25° ±5°C for a period of approximately 24 hours, after which the total resistance shall be measured. For quality conformance inspection and at the option of the supplier, the sample units may be placed in or removed from the test chamber while the chamber is at the extreme low temperature.

4.6.14 Low-temperature operation (see 3.17). The total resistance shall be measured at 25° ±5°C. The resistors shall then be placed in a chamber at 25° ±5°C and the temperature in the chamber shall be lowered to -65° ±2°C within a minimum period of 1-1/2 hours. After 1 hour ±15 minutes of stabilization at this low temperature, rated continuous working voltage (see 3.1) shall be applied for 45 minutes across the resistance element between one end terminal and the contact-arm terminal. The resistors shall then be allowed to stabilize for 1 hour ±15 minutes and the torque necessary to effect rotation of the contact arm shall

be measured. Following this measurement, the temperature shall be gradually raised to 25° ±5°C within a maximum period of 8 hours. The resistors shall then be removed from the chamber and maintained at 25° ±5°C for a period of approximately 24 hours, after which the total resistance shall be measured. For quality conformance inspection and at the option of the supplier, the sample units may be placed in or removed from the test chamber while the chamber is at the extreme high or low temperature.

4.6.15 Temperature cycling (see 3.18). Resistors shall be tested in accordance with method 102 of MIL-STD-202. The following details and exceptions shall apply:

(a) Special mounting - Resistors shall be mounted by their normal mounting means in such a manner that there is at least 1 inch of free air space around each resistor. The mounting shall be so positioned with respect to the air stream that it offers the least obstruction to the flow of air across and around the resistors.
(b) Measurements before cycling - Total resistance shall be measured as specified in 4.6.2.1.
(c) Test condition - C, except extreme high temperature shall be 175° ±2°C.
(d) Measurements after cycling - Total resistance shall be measured as specified in 4.6.2.1 and resistors shall be examined for evidence of mechanical damage.

4.6.16 High-temperature exposure (see 3.19).

4.6.16.1 Mounting. Resistors shall be mounted in such a manner as to allow electrical connections to be made to the terminals.

4.6.16.2 Procedure. The resistors shall then be exposed to an ambient temperature of 175° ±2°C. After thermal stability has been reached at 175°C, or within 6 hours after exposure, total resistance shall be measured as specified in 4.6.2.1. Following exposure for 1,000 ±48 hours, total resistance shall again be measured at 175°C. Following stabilization at room temperature, contact resistance variation shall be measured as specified in 4.6.5. Resistors shall then be examined for evidence of mechanical damage.

4.6.17 Shock, medium impact (see 3.20). Resistors shall be tested in accordance with method 205 of MIL-STD-202. The following details shall apply:

(a) Mounting - Resistors shall be mounted by their normal mounting means and affixed to an appropriate mounting fixture. A suggested mounting fixture is shown on figure 5. The mounting fixture shall be constructed in such a manner as to insure that the mounting supports remain in a static condition with reference to the shock-test table. Test leads used during this test shall be as small a wire size as practicable (e.g. AWG size 22, stranded) so that the influence of the test lead on the

6.13 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - SH, AS, OS
Air Force - 11, 17, 85

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

(Project 5905-0327)

Code "C"

APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 This appendix details the procedure for submission of samples, with related data, for qualification inspection of resistors covered by this specification. The procedure for extending qualification of the required sample to other resistors covered by this specification is also outlined herein.

20. SUBMISSION

20.1 Sample.

20.1.1 Single-type submission. A sample consisting of specimens of each type for which qualification is sought shall be submitted according to table XII. All test samples shall have 7/8 inch or 5/8 inch shaft, whichever is applicable.

20.1.2 Combined-type submission. For qualification of any bushing type, a sample consisting of the number of samples specified in table XII shall be submitted in each style and characteristic for which qualification is desired. To receive qualification for "L" or "T" bushing resistors when "N" or "S" bushing-type resistors have been submitted, an additional sample of "L" or "T" type shall be submitted as indicated in lot 5 of table XII. All test samples shall have 7/8 inch or 5/8 inch shaft, whichever is applicable for the diameter of shaft and type of bushing submitted. Six additional sample units of "C" resistor characteristic shall be subjected to group I of table VI when qualification for "C" is desired with a characteristic "B" submission.

20.2 Test data. Each submission shall be accompanied by test data covering the nondestructive tests listed in table VI which have been performed on the submitted specimens. The performance of the destructive tests by the supplier on a duplicate set of specimens is encouraged, although not required. All test data shall be submitted in duplicate.

20.3 Certification of material. When submitting samples for qualification, the supplier shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

TABLE XII. Samples for qualification inspection.

Lot	Number of specimens to be submitted (see 20.1.2)	Total resistance	Taper	Groups (see table VI) and specimen numbers							
				I	IA	II	III	IV	V	VI	VII
1	49	Lowest 1/	A	1 to 6	7 to 12	1 to 6	13 to 18	19 to 24	25 to 30	31 to 40	41 to 49
2	6	10,000 ohms	C	1 to 6	--	1 to 6	--	--	--	--	--
3	24	Critical value 2/	A	1 to 12	--	1 to 6	7 to 12	13 to 18	19 to 24	--	--
4	49	Highest 1/	A	1 to 6	7 to 12	1 to 6	13 to 18	19 to 24	25 to 30	31 to 40	41 to 49
5 3/	12 (Bushing L and T only)	Critical value 2/	A	1 to 12	--	1 to 6	--	7 to 12	--	--	--

1/For which qualification is sought.
2/See table XIII.
3/Lot 5 applies only when submitted with a standard bushing submission of lots 1 through 4.

TABLE XIII. Critical resistance values.

Style	Resistance
	Megohms
RVC5 - - - - - - - - -	.10
RVC6 - - - - - - - - -	.10

20.4 Description of items. The supplier shall submit a detailed description of the resistors being submitted for test, including a description of the case material, type of element and contacts, and methods of fastening of terminals to base and element.

30. EXTENT OF QUALIFICATION

30.1 Single-type submission. Qualification shall be for the type submitted.

30.2 Combined type submission. Qualification within a style shall be as shown in table XIV and as follows:

Between resistor characteristics:
"B" will qualify "C" when 6 additional units of "C" are subjected to group I of table VI.

Bushing types:
"N" or "S" will qualify "L" or "T" when lot 5 of table XII is submitted with a "N" or "S" (lots 1 through 4) submission.

Resistance values:
As indicated in table XII.

TABLE XIV. Extent of qualification.

Lots passing	Qualification coverage[1]
1, 2, 3, 4	All resistance values between the lowest value submitted and the highest value, all tapers.
1, 2, 4	No qualification.
1, 3, 4	All resistance values between the highest and lowest values submitted, taper-A only.
2, 3, 4	All resistance values between the critical value and the highest value submitted, all tapers.
2, 3	No qualification.
2, 4	No qualification.
1, 2, 3	Lowest value submitted to the critical value, all tapers.
1, 2, 4	No qualification.
1, 4	No qualification.
3, 4	All resistance values between the critical value and the highest value submitted, taper-A only.
1, 2	No qualification.
1, 3	All resistance values between the lowest value submitted and the critical value, taper-A only.
1	No qualification.
2 or 3 or 4	No qualification
5[2]	

1/ See 30.2 and 20.1.2 for qualification coverage between bushing types.

2/ Values and tapers qualified in lot 5 shall be the same taper and in the same resistance range qualified by lots 1 through 4 inclusive.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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